

Fertility and housing: overnight calibration

What improved and the complete target fit

Fit improved by 25.739%: the same primary-weight score fell from 26.368 to **19.581**. The selected point (initial_0506_block) passes the original gates and two fresh repeats; this does not establish an optimum.

552 search attempts: 24 censored timeout, 528 success. 6 final repeats cover 3 selected candidates. Up to 24 single-thread Torch workers; no local model computation. Model, targets, bounds and gates unchanged.

Search / reference	Loss under the same primary weights
Starting point	26.368
Equal-block weights	19.581
Relative-gap identity weights	552.788
Primary weights	22.397

All 14 rows: ten scored moments, three checks, one normalization

Moment	Target	Model	Gap	Weight	Loss
Completed fertility (normalization)	2.1	2.1	-1.66e-06	--	--
Childlessness, ages 40-44	0.198	0.201	0.003	3.55e+04	0.275
Exactly one child, among mothers	0.214	0.209	-0.004	2.70e+04	0.483
Mean age at first birth	25.976	25.933	-0.043	139.828	0.264
First births at age 30+ [V]	0.249	0.224	-0.026	0	0
Aggregate wealth / annual earnings	6.927	6.326	-0.6	7.595	2.737
Annual bequest flow / wealth	0.007	0.007	-2.34e-04	5.17e+06	0.282
Older wealth/income: p90 / median [V]	3.516	3.069	-0.447	0	0
Mean occupied rooms	5.729	5.848	0.119	128.021	1.798
Ownership, ages 30-55	0.676	0.655	-0.021	2339.362	1.076
Housing response to first birth	1.465	1.622	0.157	137.565	3.395
Rooms: 3+ vs 1-2 children at home [V]	0.385	0.353	-0.032	0	0
Recent-parent ownership gap	0.128	0.12	-0.008	2.71e+04	1.758
Children ever born at 25, capped at 3	0.81	0.535	-0.274	100	7.513

Gap = model - target. Weights and loss contributions are the common primary system. [V] denotes an untargeted validation row. Completed fertility fixes the child-benefit level and is not scored. Raw identity and block objective values are not comparable.

Economic fit: Early fertility remains 0.535 versus 0.810; wealth/earnings 6.326 versus 6.927 and ownership 0.655 versus 0.676. Relative-gap weights reach early fertility 0.657, but mean first-birth age falls to 24.047 versus 25.976.

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Fitted quantities, failure diagnosis and early-fertility tradeoffs

Ten searched parameters plus one normalized child-benefit level. Ten scored moments plus the normalization give eleven restrictions by count; local identification rank is unproven.

Parameter	Estimate	Lower	Upper	Bound / restriction
Housing supply level	6.294	0.2	80	interior
Annual discount factor	0.963	0.94	0.99	interior
Owner housing-service premium	1.094	0.1	5	interior
First-birth fixed cost	0.621	0	8	interior
First-birth taste scale	0.176	0.02	50	near lower
Later-birth taste scale	0.332	0.02	50	near lower
Bequest strength	0.125	0	8	interior
First-child housing loading	0.135	0	0.25	interior
Child-benefit curvature	0.061	0	0.8	interior
Tenure-choice taste scale	0.012	0.001	0.1	interior
Child-benefit level (normalized)	0.136	--	--	normalized; positive

Near-bound = within 1% of the full interval. All 31 parameters remain in the supporting CSV. Fixed inputs include 2% annual real interest, B15 earnings and the adopted existing-owner credit rule. No experimental credit transition is adopted or solved here.

Numerics and the earlier failures

528 search cases pass; 24 unfinished cases were stopped at the search cutoff. All six fresh repeats match the full tables and plots. Two old timeout points recover with the longer case budget. These findings do not establish equilibrium existence for every rejected parameter point.

Price-start diagnostic (reviewed): At the same failed parameter point, the original price start fails to bracket; half that start clears housing and pensions at unchanged tolerances. This identifies one numerical starting-price failure. Completed fertility is 5.063 at fixed child benefit: this is not a normalized calibration or an explanation of every earlier rejection.

How much earlier fertility can these parameters buy?

Exploratory diagnostic (reviewed): Both exact anchors pass. Halving and quartering the first-birth taste scale raise early fertility from 0.535 to 0.602 and 0.667 (target 0.810). Other-moment primary loss rises from 12.1 to 491.5 and 1398.4. At 0.667, mean first-birth age is 23.43 versus 25.98 and childlessness 0.249 versus 0.198. Of ten cases, four pass (including anchors), five time out and one violates positive child benefit. These are diagnostic probes, not an estimated global frontier.

Compare the early-fertility gap with primary-weighted loss from all other moments, excluding its own contribution. The small timing-preference probe is not a calibrated SMM estimate, global frontier, or proof of an unreachable target. Boundary proposals are diagnostics, not adoption.

Next steps and limits

First diagnose the early-fertility versus birth-timing tradeoff; retain validation rows. Test a robust price-bracketing strategy without weakening gates. Examine occupied-state policy behavior and local identification before any further production calibration.

Buyer-conditional standard plots do not certify every occupied stayer policy. High-wealth housing/ownership downturns and retirement behavior remain caveats. First-birth housing uses a four-year model contrast; estate and empirical child-transfer measures also differ. Full tables and the stable 17 plots are separate supporting artifacts.